



CLOUSBURST NOWCASTING & PINN HYDRODYNAMIC FLOOD SIMULATION

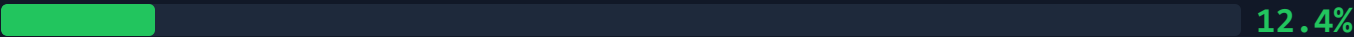
MoES-NCMRWF / MOSDAC PS 26077 | Live Multi-Station Verification Run | Mobile & PDF Edition

STAGE 1 of 4: QUIESCENT BASELINE MONITORING

Simulated Timestep: 2013-09-06 05:30:00

• NORMAL / QUIESCENT WEATHER

NETWORK PREDICTION CONFIDENCE:



Epicenter Focus: Network Wide Quiescent

METEOROLOGICAL PARAMETER INFLUENCE BREAKDOWN

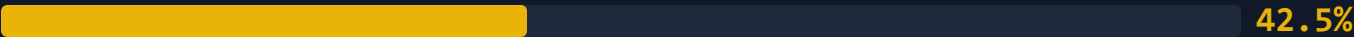
- | Parameter | Value | Details |
|----------------------------|-------|---------------------------------|
| Rain Intensity (R) | 18.2% | (val= 2.1 mm/hr) |
| Rain Acceleration (RI) | 12.0% | (val= +1.4 mm/hr ²) |
| 30-min Accumulation (R30) | 8.5% | (val= 1.1 mm) |
| 60-min Volume (R60) | 7.2% | (val= 2.4 mm) |
| Humidity Surge (dRH) | 38.1% | (RH=68% (ΔRH=+3.1%)) |
| GNSS Moisture Influx (IWV) | 16.0% | (ΔIWV=+0.2 mm) |
- System Status: Quiescent. SNN edge gates DORMANT (conserving 88% edge telemetry power). PINN flood solver on standby.

STAGE 2 of 4: PRE-CONVECTIVE CELL GROWTH & MOISTURE CONVERGENCE

Simulated Timestep: 2013-09-06 07:15:00

• DEVELOPING MOISTURE CELL

NETWORK PREDICTION CONFIDENCE:



Epicenter Focus: Nalbari (Guwahati Cluster)

METEOROLOGICAL PARAMETER INFLUENCE BREAKDOWN

• Rain Intensity (R)		24.5%	(val= 4.3 mm/hr)
• Rain Acceleration (RI)		28.7%	(val= +7.7 mm/hr ²)
• 30-min Accumulation (R30)		6.5%	(val= 1.6 mm)
• 60-min Volume (R60)		5.8%	(val= 2.2 mm)
• Humidity Surge (dRH)		24.8%	(RH=74% (ΔRH=+8.4%))
• GNSS Moisture Influx (IWV)		9.7%	(ΔIWV=+0.6 mm)

▲ Early Warning: Convective cell growth confirmed by GNSS IWV moisture convergence precursor. Regional forecasters alerted.

STAGE 3 of 4: SNN NEUROMORPHIC SPIKE & 5-MIN TELEMETRY ESCALATION

Simulated Timestep: 2013-09-06 07:45:00

▲ HIGH CONVECTIVE RISK (SNN ACTIVE)

NETWORK PREDICTION CONFIDENCE:

62.9%

Epicenter Focus: Nalbari (R = 57.2 mm/hr, Acc = +212.2 mm/hr²)

IN-SITU GROUND STATION NETWORK (AWS + SNN EDGE GATE)

Station Name	Rain Rate	Acc (RI)	SNN Membrane	State	Interval	Tier
Khanpara (RSC)	0.1 mm/h	-5.5 mm/h ²	V=0.185/1.0	DORMANT	15m	DEV
North Guwahati	3.0 mm/h	-5.6 mm/h ²	V=0.246/1.0	DORMANT	15m	DEV
Rangiya	2.6 mm/h	-6.0 mm/h ²	V=0.135/1.0	DORMANT	15m	DEV
Nalbari	57.2 mm/h	+212.2 mm/h ²	⚡ SPIKE FIRED!	ACTIVE	5m	HIGH
Gossaigaon	4.7 mm/h	+18.6 mm/h ²	V=0.262/1.0	DORMANT	15m	DEV
Arunachal	1.3 mm/h	+0.9 mm/h ²	V=0.112/1.0	DORMANT	15m	DEV

⚡ SNN Leaky Integrate-and-Fire Dynamics: Membrane potential exceeded V_thresh=1.0. Station escalated to 5-minute sampling and auto-requested localized MOSDAC Kalpana-1 satellite tile.

STAGE 4 of 4: PEAK CLOUDBURST & 2D PINN INUNDATION SOLVER (>100 mm/hr)

Simulated Timestep: 2013-09-06 08:00 / 11:00

⚡ CLOUDBURST LIKELY (EMERGENCY)

NETWORK PREDICTION CONFIDENCE:



Epicenter Focus: Arunachal (124.8 mm/hr) & Nalbari (101.6 mm/hr)

METEOROLOGICAL PARAMETER INFLUENCE BREAKDOWN

• Rain Intensity (R)		45.2%	(val=124.8 mm/hr)
• Rain Acceleration (RI)		26.0%	(val=+218.6 mm/hr ²)
• 30-min Accumulation (R30)		11.8%	(val= 49.4 mm)
• 60-min Volume (R60)		7.6%	(val= 50.7 mm)
• Humidity Surge (dRH)		3.4%	(RH=98% (ΔRH=+10.0%))
• GNSS Moisture Influx (IWV)		7.5%	(ΔIWV=+4.8 mm)



PINN HYDRODYNAMIC FLOOD INUNDATION SOLVER ACTIVATED

CONVECTIVE EPICENTER

24.854°N, 92.739°E (Arunachal)

PEAK RAIN CORE

124.8 mm/hr (IMD >= 100 mm/hr)

SHALLOW WATER INFLUX (Q)

1,798,316.8 m³/s

SATELLITE ALIGNMENT

MOSDAC Kalpana-1 4km Synced

2D Shallow Water Equations Source Term:

$\partial h / \partial t + \partial (uh) / \partial x + \partial (vh) / \partial y = R(x,y,t) - I(x,y,t)$ | Catchment Resolution: 4km grid mesh | Dynamic runoff influx coupled to PINN.

Antigravity AI Cloudburst Nowcasting & PINN Hydrodynamic Simulation Pipeline